Sentences should also be grammatically correct and unambiguous. Difficult words should be avoided and their simple synonyms should be used.

3.7 Software Requirements Specification Document

After requirements elicitation, we prepare the IRD. This document gives an overview of the system. Now a detailed document is prepared on the basis of the IRD and is called *software requirements specification* (SRS) document. This document is used as a legal document which acts as a contract between customers and developers. On the basis of this document, the developers know what to build, and the customers know what to expect, and this document may also be used to validate that the built system satisfies the requirements. We use IEEE recommended practice for software requirements specifications—IEEE standard 830-1998 for preparing the SRS document.

3.7.1 Nature of the SRS Document

The following issues shall be addressed by the SRS writer:

1. *Functionality:* What functions the software is supposed to perform?
2. *External interfaces:* With what all external entities does the system interact? This may include the number of users, response time, recovery time, processing time, etc.
3. *Performance:* How does the software address performance issues? This may include people, hardware, external databases, etc.
4. *Quality attributes:* What are the considerations for non-functional requirements? These may include availability, correctness, maintainability, portability, reliability, security, testability, etc. These non-functional requirements should also be properly placed in the SRS document.
5. *Design constraints imposed on implementation:* All constraints should be highlighted which have an impact on implementation. This may include limitations of the operating environment, programming language, database, resources, policies for database integrity, etc.

The SRS writer(s) should not include design and implementation details. It should be written in a simple, clear and unambiguous language which may be understandable to all developers and customers.

3.7.2 Organization of the SRS Document

We have used IEEE recommended practice for SRS which is known as IEEE Std. 830-1998. The standard was approved on June 25, 1998. The document is used to specify requirements of the software which is to be developed. It helps the customers to know what they want from the proposed software system. It also helps the developer to understand exactly what the customer wants. There are four sections of the SRS document and are given in Table 3.8 (IEEE, 1998). The first two sections are the same for all projects, but Section 3 provides special tailoring as mentioned "specific requirements" for different projects.

Table 3.8 SRS document outline as per IEEE Std 830-1998

1.	Introduction 1.1 Purpose 1.2 Scope 1.3 Definitions, acronyms and abbreviations 1.4 References 1.5 Overview
2.	Overall description 2.1 Product perspective 2.1.1 System interfaces 2.1.2 User interfaces 2.1.3 Hardware interfaces 2.1.4 Software interfaces 2.1.5 Communications interfaces 2.1.6 Memory constraints 2.1.7 Operations 2.1.8 Site adaptation requirements 2.2 Product functions 2.3 User characteristics 2.4 Constraints 2.5 Assumptions and dependencies 2.6 Apportioning of requirements
3.	Specific requirements 3.1 External interfaces 3.2 Functions 3.3 Performance requirements 3.4 Logical database requirements 3.5 Design constraints 3.5.1 Standards compliance 3.6 Software system attributes 3.6.1 Reliability 3.6.2 Availability 3.6.3 Security 3.6.4 Maintainability 3.6.5 Portability 3.7 Organizing the specific requirements 3.7.1 System mode 3.7.2 User class 3.7.3 Objects 3.7.4 Feature 3.7.5 Stimulus 3.7.6 Response 3.7.7 Functional hierarchy 3.8 Additional comments
4.	Supporting information

These sections are discussed below along with their section numbers as in Table 3.8. The concepts are also explained with the help of the case study of the LMS. The complete SRS is given in the Appendix.

1. Introduction

This section gives the overview of the complete SRS document.

1.1 Purpose

We should describe the purpose of the SRS document and also list the intended audience for the document. The purpose of the LMS is given as:

> **1.1 Purpose**
>
> The library management system (LMS) maintains the information about various books available in the library. The LMS also maintains records of all the students, faculty and employees in the university. These records are maintained in order to provide membership to them.

1.2 Scope

The task of this subsection is to:

(i) Assign a name to the software under development.
(ii) List the functions which will be provided and also those functions which the software will not do.
(iii) Explain the applications of the software along with possible benefits and objectives.
(iv) Maintain consistency amongst statements.

The scope of the LMS is given as:

> **1.2 Scope**
>
> The name of the software is library management system (LMS). The system will be referred to as LMS in rest of the SRS. The proposed LMS must be able to perform the following functions:
>
> **Dos**
>
> 1. Issue of login ID and password to system operators.
> 2. Maintain details of books available in the library.
> 3. Maintain details of the students in the university to provide student membership.
> 4. Maintain details of the faculty and employees in the university to provide faculty and employees membership.
> 5. Issue a book.
> 6. Bring a book back, and calculates fine if the book is being returned after the specified due date.
> 7. Reserve a book.
> 8. Provide search facility to check the availability of a book in the library.
> 9. Generate the following reports:
>   (a) List of books issued by a member.
>   (b) Details of books issued and returned on daily basis.
>   (c) Receipt of fine.

(d) Total available books in the library.
(e) List of library members along with issued books.
(f) List of available books in the library:
- Author-wise
- Book title-wise
- Publisher-wise
- Subject-wise

Don'ts
1. Periodicals section is not covered.
2. Book bank (if any) is not covered.
3. Records of digital books are not available.
4. Purchase of books facility is not available.

Benefits
The LMS provides the following benefits:
1. Easy searching of books.
2. Efficient issue and return of books.
3. Accurate and automated fine calculation.
4. Printing of reports.

1.3 Definitions, Acronyms and Abbreviations

Define all terms, acronyms and abbreviations. This may help to make the SRS document more readable, understandable and clear to all the stakeholders. The definitions and acronyms used in the LMS are given as:

1.3 Definitions, Acronyms and Abbreviations

SRS: Software requirement specification

LMS: Library management system

System operator: System administrator, library staff, data entry operator

RAM: Random access memory

Accession number: It is a unique sequence number allocated to each book in the catalog.

Student: Any candidate admitted in a programme offered by a school.

System administrator/Administrator: User having all the privileges to operate the LMS.

Data entry operator (DEO): User having privileges to maintain book, student and faculty/employee details.

Library staff: User having privilege to issue, return, reserve and query books.

Faculty: Teaching staff of the university—Professor, Associate Professor, Assistant Professor

School: Academic unit that offers various programmes

1.4 References

This subsection provide a list of all documents including books and research papers referenced anywhere in the SRS document. The referenced material used in LMS is given as:

> **1.4 References**
>
> (a) *Software Engineering* by K.K. Aggarwal & Yogesh Singh, New Age Publishing House, 3rd Edition, 2008.
>
> (b) IEEE Recommended Practice for Software Requirements Specifications— IEEE Std 830-1998.
>
> (c) IEEE Standard for Software Test Documentation—IEEE Std. 829-1998.

1.5 Overview

This subsection of the SRS gives the overview of the software system. Explain what the rest of the SRS contains and also gives details about the organization of the SRS document.

2. Overall Description

This section discusses the general factors which affect the requirements, and the final product specific requirements are not discussed. This section may help to write specific requirements of section 3 of the SRS document. The overview of the LMS is given as:

> **2. Overall Description**
>
> The LMS maintains records of books in the library and membership details of students, faculty and employees in the university. It is assumed that approved books have already been purchased by the acquisition section. It is also assumed that the student has already been admitted in the university for a specific programme. The system administrator will receive lists of the admitted students (school-wise and programme-wise) from the academic section. The establishment section will provide the list of the faculty and employees appointed in the university.
>
> The LMS issues books to students, faculty and employees and brings the same back from them. The LMS generates fine if the book is returned by a student after the due date. It also allows students, faculty and employee to reserve a book. The student/faculty/employee can access LMS on the university's library LAN in order to search the availability of the book from the library.
>
> The administrator/DEO will have to maintain the following information:
>
> - Book details
> - Student membership details
> - Faculty and employee membership details
>
> The administrator/library staff will perform the following functions:
>
> - Issue a book
> - Return a book
> - Reserve a book
> - Query a book

- Generate reports
    - List of books issued to a member
    - Details of books issued and returned on daily basis
    - Receipt of fine
    - Total available books in the library
    - List of library members along with issued books

The administrator/student/faculty/employee requires the following information from the system:

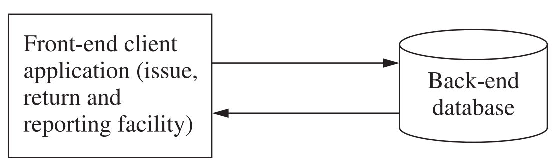
- List of available books in the library:
    - Author-wise
    - Book title-wise
    - Publisher-wise
    - Subject-wise

2.1 Product Perspective

This subsection puts the product into perspective with other related products. We should specify the environment which may be web-enabled, stand-alone system, or usage of local area network. A block diagram showing the major components of the large system, interconnections and external interfaces can be helpful. The product perspective of LMS is given as:

2.1 Product Perspective

The LMS shall be developed using client/server architecture and will be compatible with Microsoft Windows Operating System. The front-end of the system will be developed using Visual Basic 6.0 and the back-end will be developed using MS SQL Server 2005.



2.1.1 System Interfaces

System interfaces are addressed which are required by the product during its usage.

2.1.2 User Interfaces

This subsection should specify:

(i) Logical characteristics of each interface between the product and its users.
(ii) How the system can be best used by the user. A list of dos and don'ts for the user may also be specified.

The user interfaces of the LMS are given as:

2.1.2 User Interfaces

The LMS will have the following user-friendly and menu-driven interfaces:

(a) **Login:** to allow the entry of only authorized users through valid login ID and password.

(b) **Book details:** to maintain book details.

(c) **Student membership details:** to maintain student membership details.

(d) **Faculty/employee membership details:** to maintain faculty/employee membership details.

(e) **Issue book:** to allow library staff to issue a book from the library.

(f) **Return book:** to allow library staff to bring a book back to the library.

(g) **Reserve book:** to allow library staff to reserve a book.

(h) **Query book:** to check availability of a book in the library.

The software should generate the following information:

(a) Details of books issued and returned on daily basis.
(b) Details of books available in the library.
(c) Details of books issued to a student.
(d) List of library members with issued books.
(e) Receipt of fine

2.1.3 Hardware Interfaces

Logical characteristics of each interface between the software product and the hardware component is to be specified. For example, issues such as screen resolution, support for printer, single user/multi-user system, etc. should be addressed. These issues for the LMS are shown as follows:

2.1.3 Hardware Interfaces

(a) Screen resolution of at least 640 × 480 or above.

(b) Support for printer (dot matrix, deskjet, laserjet).

(c) Computer systems will be in the networked environment as it is a multi-user system.

2.1.4 Software Interfaces

This subsection should specify the interface of the software product with other software applications. For example, operating system, front end tools, back end tools, the specification and version number of the software, etc. should also be specified. The software interfaces for the LMS are given as:

2.1.4 Software Interfaces

(a) MS-Windows Operating System (NT/XP/Vista)

(b) Microsoft Visual Basic 6.0 for designing front-end

(c) MS SQL Server 2005 for back-end

2.1.5 Communication Interfaces

This subsection specifies the various interfaces to communication such as local network, protocol, etc. For example, whether the software product will use LAN, web-enabled services or stand-alone system. In the LMS, communication is via local area network (LAN).

2.1.6 Memory Constraints

Any constrains related to primary and secondary memories are specified. For example, for the LMS, we may specify "At least 512 MB RAM and 500 MB space of hard disk will be required to run the software".

2.1.7 Operations

Various operations in the user's organization are specified. This may include data processing support functions, backup and recovery operations.

2.1.8 Site Adaptation Requirements

Any specific requirements related to the hardware and software interfaces at site are to be specified in this subsection. In the LMS, the terminal at the client site will have to support the hardware and software interfaces specified in sections 2.1.3 and 2.1.4, respectively.

2.2 Product Functions

This section should list the summary of all functions that the software shall perform. Functions should be listed in a simple and clear language and must be understandable to the customer. The product functions for the LMS are given as:

> **2.2 Product Functions**
>
> The LMS will allow access only to the authorized users with specific roles (system administrator, library staff, DEO and student). Depending upon the user's role, he/she will be able to access only specific modules of the system.
>
> A summary of major functions that the LMS shall perform includes:
>
> - A login facility for enabling only authorized access to the system.
> - The system administrator/DEO will be able to add, modify, delete or view book, student, faculty/employee and login information.
> - The system administrator/library staff will be able to issue, bring back, and reserve book.
> - The system administrator/library staff will be able to query a book in order to check the availability of the book in the library.
> - The system administrator/students/faculty/employee will be able to search a book from the library catalogue (author-wise, title-wise and publisher-wise).
> - The system administrator/library staff will be able to generate various reports from the LMS.

2.3 User Characteristics

Characteristics of users in terms of their qualification, skills sets and experience should be specified in this subsection. The user characteristics for the LMS are given as:

2.3 User Characteristics

Qualification: At least matriculation and comfortable with English.

Experience: Should be well versed/informed about the processes of the university library.

Technical experience: Elementary knowledge of computers.

2.4 Constraints

We may list constraints that may limit the developer's implementation options. These constraints may include:

(i) Regulatory policies
(ii) Hardware limitations
(iii) Interface to other applications
(iv) Parallel operations
(v) Audit functions
(vi) Control functions
(vii) Higher-order language requirements
(viii) Protocols
(ix) Reliability requirements
(x) Safety, security and criticality of the application

The constraints for the LMS are given as:

2.4 Constraints

- The software does not maintain records of periodicals.
- There will be only one administrator.
- The delete operation is available to the administrator and DEO. To reduce the complexity of the system, there is no check on delete operation. Hence, the administrator/DEO should be very careful before deletion of any record and he/she will be responsible for data consistency.
- The user will not be allowed to update the primary key.

2.5 Assumptions and Dependencies

This subsection includes assumptions and dependencies for the implementation of the software product. These assumptions may be very important and may become input for the design and implementation of the software system.

2.5 Assumptions and Dependencies

- The acquisition section of the library will provide the list of the books purchased by the library.
- The academic section will provide the list of the admitted students (school-wise and programme-wise).
- The establishment section will provide the list of the faculty members in various schools of the university. It will also provide the list of employees working in various branches such as examination, academics, establishment and schools of the university.
- The login ID and password must be created by the system administrator and communicated to the concerned user confidentially to avoid unauthorized access to the system.

2.6 Apportioning of Requirements

This subsection list those requirements that can be delayed and may be incorporated in the future version of the software system. For the LMS, there is no such requirement.

3. Specific Requirements

All requirements stated in section 2 must be written in detail and at a level sufficient for the designers to understand and design the system. The testers should test that the requirements are met by the system. All requirements should conform to the guidelines of good requirements as discussed in Section 3.6. Some of the characteristics of good requirements are correctness, completeness, unambiguity, consistence, verifiability, modifiability, traceability, etc.

In addition to the basic format given by IEEE 830-1998, we may also construct Section 3 according to any one of the following additional templates given in Section 3.7 of the SRS.

System Mode

Depending on the mode of operation, we may choose either of the two outlines given in Tables 3.9 and 3.10.

Table 3.9 Template of SRS section 3 organized by mode

3. Specific requirements
3.1 External interface requirements 3.1.1 User interfaces 3.1.2 Hardware interfaces 3.1.3 Software interfaces 3.1.4 Communications interfaces
3.2 Functional requirements 3.2.1 Mode 1 3.2.1.1 Functional requirement 1.1 . . . 3.2.1.*n* Functional requirement 1.*n* 3.2.2 Mode 2 . . . 3.2.*m* Mode *m* 3.2.*m*.1 Functional requirement *m*.1 . . . 3.2.*m*.*n* Functional requirement *m*.*n*
3.3 Performance requirements
3.4 Design constraints
3.5 Software system attributes
3.6 Other requirements

Table 3.10 Another template of SRS section 3 organized by mode

3. Specific requirements
3.1 Functional requirements
3.1.1 Mode 1
3.1.1.1 External interfaces
3.1.1.1.1 User interfaces
3.1.1.1.2 Hardware interfaces
3.1.1.1.3 Software interfaces
3.1.1.1.4 Communications interfaces
3.1.1.2 Functional requirements
3.1.1.2.1 Functional requirement 1
.
.
.
3.1.1.2.*n* Functional requirement *n*
3.1.1.3 Performance
3.1.2 Mode 2
.
.
.
3.1.*m* Mode *m*
3.2 Design constraints
3.3 Software system attributes
3.4 Other requirements

User Class

Some systems provide different sets of functions to different classes of users. The outline is given in Table 3.11.

Table 3.11 Template of SRS section 3 organized by user class

3. Specific requirements
3.1 External interface requirements
3.1.1 User interfaces
3.1.2 Hardware interfaces
3.1.3 Software interfaces
3.1.4 Communications interfaces
3.2 Functional requirements
3.2.1 User class 1
3.2.1.1 Functional requirement 1.1
.
.
.
3.2.1.*n* Functional requirement 1.*n*

(*Contd.*)

Table 3.11 Template of SRS section 3 organized by user class (*Contd.*)

3.2.2 User class 2 . . . 3.2.*m* User class *m* 3.2.*m*.1 Functional requirement *m*.1 . . . 3.2.*m*.*n* Functional requirement *m*.*n*
3.3 Performance requirements
3.4 Design constraints
3.5 Software system attributes
3.6 Other requirements

Objects

The outline of arranging section 3 by objects is given in Table 3.12.

Table 3.12 Template of SRS section 3 organized by object

3. Specific requirements
3.1 External interface requirements 3.1.1 User interfaces 3.1.2 Hardware interfaces 3.1.3 Software interfaces 3.1.4 Communications interfaces
3.2 Classes/Objects 3.2.1 Class/Object 1 3.2.1.1 Attributes (direct or inherited) 3.2.1.1.1 Attribute 1 . . . 3.2.1.1.*n* Attribute *n* 3.2.1.2 Functions (services, methods, direct or inherited) 3.2.1.2.1 Functional requirement 1.1 . . . 3.2.1.2.*m* Functional requirement 1.*m* 3.2.1.3 Messages (communications received or sent) 3.2.2 Class/Object 2 . . . 3.2.*p* Class/Object *p*
3.3 Performance requirements
3.4 Design constraints
3.5 Software system attributes
3.6 Other requirements

Feature

A feature is an externally desired service by the system that may require a sequence of inputs to affect the desired result. Each feature is generally described as a sequence of stimulus response pairs. The outline is given in Table 3.13.

Table 3.13 Template of SRS section 3 organized by feature

3. Specific requirements
3.1 External interface requirements 3.1.1 User interfaces 3.1.2 Hardware interfaces 3.1.3 Software interfaces 3.1.4 Communications interfaces
3.2 System features 3.2.1 System feature 1 3.2.1.1 Introduction/Purpose of feature 3.2.1.2 Stimulus/Response sequence 3.2.1.3 Associated functional requirements 3.2.1.3.1 Functional requirement 1 . . . 3.2.1.3.*n* Functional requirement *n* 3.2.2 System feature 2 . . . 3.2.*m* System feature *m* . . .
3.3 Performance requirements
3.4 Design constraints
3.5 Software system attributes
3.6 Other requirements

Stimulus

Some systems are best organized by describing their functions in terms of stimuli. The outline is given in Table 3.14.

Table 3.14 Template of SRS section 3 organized by stimulus

3. Specific requirements
3.1 External interface requirements 3.1.1 User interfaces 3.1.2 Hardware interfaces 3.1.3 Software interfaces 3.1.4 Communications interfaces

(*Contd.*)

Table 3.14 Template of SRS section 3 organized by stimulus (*Contd.*)

3.2 Functional requirements 3.2.1 Stimulus 1 3.2.1.1 Functional requirement 1.1 . . . 3.2.1.*n* Functional requirement 1.*n* 3.2.2 Stimulus 2 . . . 3.2.*m* Stimulus *m* 3.2.*m*.1 Functional requirement *m*.1 . . . 3.2.*m*.*n* Functional requirement *m*.*n*
3.3 Performance requirements
3.4 Design constraints
3.5 Software system attributes
3.6 Other requirements

Response

Some systems are best organized by describing their functions in support of the generation of a response. The outline is the same as given in Table 3.14 (stimulus is replaced by response).

Functional Hierarchy

Functions are organized in terms of their hierarchy with respect to either common inputs, common outputs or common internal data access. Data flow diagrams and data dictionaries may be used to show the relationships amongst the functions and data. The outline is given in Table 3.15.

Table 3.15 Template of SRS section 3 organized by functional hierarchy

3. Specific requirements
3.1 External interface requirements 3.1.1 User interfaces 3.1.2 Hardware interfaces 3.1.3 Software interfaces 3.1.4 Communications interfaces
3.2 Functional requirements 3.2.1 Information flows 3.2.1.1 Data flow diagram 1 3.2.1.1.1 Data entities 3.2.1.1.2 Pertinent processes 3.2.1.1.3 Topology

(*Contd.*)

Table 3.15 Template of SRS section 3 organized by functional hierarchy (*Contd.*)

	3.2.1.2	Data flow diagram 2	
		3.2.1.2.1	Data entities
		3.2.1.2.2	Pertinent processes
		3.2.1.2.3	Topology
	.		
	.		
	.		
	3.2.1.*n*	Data flow diagram *n*	
		3.2.1.*n*.1	Data entities
		3.2.1.*n*.2	Pertinent processes
		3.2.1.*n*.3	Topology
3.2.2	Process descriptions		
	3.2.2.1	Process 1	
		3.2.2.1.1	Input data entities
		3.2.2.1.2	Algorithm or formula of process
		3.2.2.1.3	Affected data entities
	3.2.2.2	Process 2	
		3.2.2.2.1	Input data entities
		3.2.2.2.2	Algorithm or formula of process
		3.2.2.2.3	Affected data entities
	.		
	.		
	.		
	3.2.2.*m*	Process *m*	
		3.2.2.*m*.1	Input data entities
		3.2.2.*m*.2	Algorithm or formula of process
		3.2.2.*m*.3	Affected data entities
3.2.3	Data construct specifications		
	3.2.3.1	Construct 1	
		3.2.3.1.1	Record type
		3.2.3.1.2	Constituent fields
	3.2.3.2	Construct 2	
		3.2.3.2.1	Record type
		3.2.3.2.2	Constituent fields
	.		
	.		
	.		
	3.2.3.*p*	Construct *p*	
		3.2.3.*p*.1	Record type
		3.2.3.*p*.2	Constituent fields
3.2.4	Data dictionary		
	3.2.4.1	Data element 1	
		3.2.4.1.1	Name
		3.2.4.1.2	Representation
		3.2.4.1.3	Units/Format
		3.2.4.1.4	Precision/Accuracy
		3.2.4.1.5	Range
	3.2.4.2	Data element 2	
		3.2.4.2.1	Name
		3.2.4.2.2	Representation

(*Contd.*)

Table 3.15 Template of SRS section 3 organized by functional hierarchy (*Contd.*)

3.2.4.2.3 Units/Format 3.2.4.2.4 Precision/Accuracy 3.2.4.2.5 Range . . . 3.2.4.*q* Data element *q* 3.2.4.*q*.1 Name 3.2.4.*q*.2 Representation 3.2.4.*q*.3 Units/Format 3.2.4.*q*.4 Precision/Accuracy 3.2.4.*q*.5 Range
3.3 Performance requirements
3.4 Design constraints
3.5 Software system attributes
3.6 Other requirements

3.1 External Interfaces

External interfaces provide mechanisms to enter data into the system. Sometimes, we may also specify the outputs from the software system. We may use different types of forms for the entry of the data. A few forms of the LMS are given that are used to enter data into the LMS. Formats for various fields are also defined.

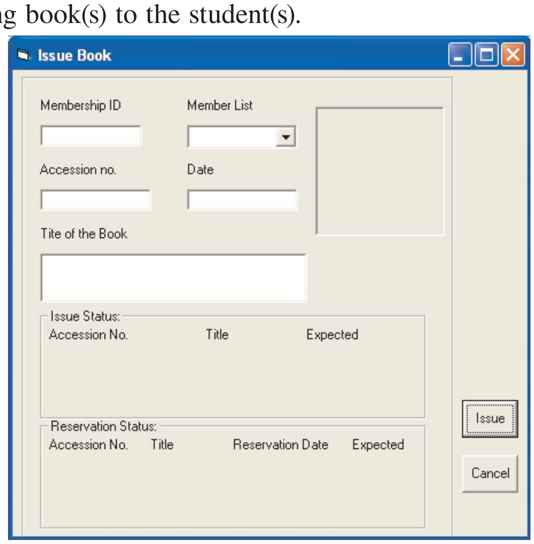
The specific requirements for issuing book in the LMS following the basic format of SRS given in IEEE Std 830-1998 are given as follows:

3. Specific Requirements

This section contains the software requirements in detail along with the various forms to be developed.

Issue Book

This form will be accessible only to the system administrator and library staff. It will allow him/her to issue existing book(s) to the student(s).



Various fields available on this form will be:

- **Membership ID:** Numeric and will have value from 100 to 19999. The system shall not allow the user to enter non-numeric characters and out of range values.
- **Accession number:** Numeric and will have value from 10 to 199999. The system shall not allow the user to enter non-numeric characters and out of range values.
- **Title:** Alphanumeric of length 3 to 100 characters. Special characters (except brackets) are not allowed. Numeric data will be allowed. The system shall not allow the user to enter special characters and out of range values.
- Issue status (the following member information will be displayed):
  - ♦ **Accession number:** Numeric and will have value from 10 to 199999. The system shall not allow the user to enter non-numeric characters and out of range values.
  - ♦ **Title:** Alphanumeric of length 3 to 100 characters. Special characters (except brackets) are not allowed. Numeric data will be allowed. The system shall not allow the user to enter special characters and out of range values.
  - ♦ **Expected:** Date of return. It will be of mm/dd/yyyy format and will have 10 alphanumeric characters. The system shall not allow the user to enter any other format.
- Reservation status (the following fields will be displayed):
  - ♦ **Accession number:** Numeric and will have value from 10 to 199999. The system shall not allow the user to enter non-numeric characters and out of range values.
  - ♦ **Title:** Alphanumeric of length 3 to 100 characters. Special characters (except brackets) are not allowed. Numeric data will be allowed. The system shall not allow the user to enter special characters and out of range values.
  - ♦ **Reservation date:** It will be of mm/dd/yyyy format. It will have 10 alphanumeric characters. The system shall not allow the user to enter any other format.
  - ♦ **Expected:** Date of return. It will be of mm/dd/yyyy format and will have 10 alphanumeric characters. The system shall not allow the user to enter any other format.

3.2 Functions

Functional requirements define the fundamental actions that must take place in the software when inputs are accepted, and processed, and outputs are accordingly generated. Generally, the most popular way to implement this process is to write use case description. The use case description addresses all actions when an actor interacts with the system for a specified purpose. We may also include validity checks on inputs, sequencing information about operations, and error handling/response to abnormal situations. A few use case descriptions of LMS are given below in the specified template. The validity check and error handling information is also provided.

MAINTAIN BOOK DETAILS

A. Use Case Description

Introduction

This use case documents the steps that the administrator/DEO must follow in order to maintain book details and add, update, delete and view book information.

Actors

Administrator
Data entry operator

Precondition

The administrator/DEO must be logged onto the system before this use case begins.

Postcondition

If the use case is successful, then the book information is added, updated, deleted or viewed. Otherwise, the system state is unchanged.

Flow of Events

Basic Flow

This use case starts when the administrator/DEO wishes to add/update/delete/view book information.

1. The system requests that the administrator/DEO specify the function he/she would like to perform (either add a book, update a book, delete a book or view a book).
2. Once the administrator/DEO provides the requested information, one of the subflows is executed.
   - If the administrator/DEO selects "Add a Book", the **Add a Book** subflow is executed.
   - If the administrator/DEO selects "Update a Book", the **Update a Book** subflow is executed.
   - If the administrator/DEO selects "Delete a Book", the **Delete a Book** subflow is executed.
   - If the administrator/DEO selects "View a Book", the **View a Book** subflow is executed.

Basic Flow 1: Add a Book

The system requests that the administrator/DEO enter the book information. This includes:

- Accession number (book barcode ID)
- Subject descriptor
- ISBN
- Title
- Language
- Author
- Publisher

Once the administrator/DEO provides the requested information, the book is added to the system.

Basic Flow 2: Update a Book

1. The system requests that the administrator/DEO enter the accession number.

2. The administrator/DEO enters the accession number.
3. The system retrieves and displays the book information.
4. The administrator/DEO makes the desired changes to the book information. This includes any of the information specified in the **Add a Book** subflow.
5. Once the administrator/DEO updates the necessary information, the system updates the book information with the updated information.

Basic Flow 3: Delete a Book

1. The system requests that the administrator/DEO specify the accession number.
2. The administrator/DEO enters the accession number. The system retrieves and displays the required information.
3. The system prompts the administrator/DEO to confirm the deletion of the book record.
4. The administrator/DEO verifies the deletion.
5. The system deletes the record.

Basic Flow 4: View a Book

1. The system requests that the administrator/DEO specify the accession number.
2. The system retrieves and displays the book information.

Alternative Flows

Alternative Flow 1: Invalid Entry

If in the **Add a Book** or **Update a Book** flow, the actor enters invalid accession number/ISBN/title/author/publisher/language/subject descriptor or leaves the accession number/ISBN/title/author/publisher/language/subject descriptor empty, the system displays an appropriate error message. The actor returns to the basic flow and may reenter the invalid entry.

Alternative Flow 2: Book Already Exists

If in the **Add a Book** flow, a book with a specified accession number already exists, the system displays an error message. The administrator returns to the basic flow and may reenter the book.

Alternative Flow 3: Book Not Found

If in the **Update a Book** or **Delete a Book** or **View a Book** flow, the book information with the specified accession number does not exist, the system displays an error message. The administrator returns to the basic flow and may reenter the accession number.

Alternative Flow 4: Update Cancelled

If in the **Update a Book** flow, the administrator/DEO decides not to update the book, the update is cancelled and the **Basic Flow** is restarted at the beginning.

Alternative Flow 5: Delete Cancelled

If in the **Delete a Book** flow, the administrator/DEO decides not to delete the book, the delete is cancelled and the **Basic Flow** is restarted at the beginning.

Alternative Flow 6: Deletion Not Allowed

If in the **Delete a Book** flow, issue/return/reserve details of the book selected exist, then the system displays an error message. The administrator returns to the basic flow and may reenter the student membership number.

Alternative Flow 7: User Exits

This allows the user to exit at any time during the use case. The use case ends.

Special Requirements
None

Associated Use Cases
Login

B. Validity Checks

(i) Only the administrator/DEO will be authorized to access the Book Details module.
(ii) Every book will have a unique accession number.
(iii) Accession number cannot be blank.
(iv) Accession number can only have value from 10 to 199999 digits.
(v) Subject descriptor cannot be blank.
(vi) ISBN number cannot be blank.
(vii) Length of ISBN number for any user can only be equal to 11 digits.
(viii) ISBN number will not accept alphabets, special characters and blank spaces.
(ix) Title cannot be blank.
(x) Length of title can be of 3 to 100 characters.
(xi) Title will only accept alphabetic characters, brackets, numeric digits and blank spaces.
(xii) Language cannot be blank.
(xiii) Author (first name and last name) cannot be blank.
  (a) Length of first name and last name can be of 3 to 100 characters.
  (b) First name and last name will not accept special characters and numeric digits.
(xiv) Publisher cannot be blank.
(xv) Length of first name and last name can be of 3 to 300 characters.
(xvi) Publisher will not accept special characters and numeric digits.

C. Sequencing Information
None

D. Error Handling/Response to Abnormal Situations
If any of the validation flows does not hold true, an appropriate error message will be prompted to the user for doing the needful.

MAINTAIN STUDENT MEMBERSHIP DETAILS

A. Use Case Description

Introduction
This use case documents the steps that the Administrator/DEO must follow in order to maintain student membership details. This includes adding, updating, deleting and viewing student information.

Actors
Administrator
DEO

Precondition
The administrator/DEO must be logged onto the system before this use case begins.

Postcondition

If the use case is successful, then student information is added/updated/deleted/viewed from the system. Otherwise, the system state is unchanged.

Flow of Events

Basic Flow

This use case starts when the administrator/DEO wishes to add, update, delete or view student information from the system.

1. The system requests that the administrator/DEO specify the function he/she would like to perform (either add a student, update a student record, delete a student record).
2. Once the administrator/DEO provides the requested information, one of the subflows is executed.
   - If the administrator/DEO selects "Add a Student", the **Add a Student** subflow is executed.
   - If the administrator/DEO selects "Update a Student", the **Update a Student** subflow is executed.
   - If the administrator/DEO selects "Delete a Student", the **Delete a Student** subflow is executed.
   - If the administrator/DEO selects "View a Student", the **View a Student** subflow is executed.

Basic Flow 1: Add a Student

The system requests that the administrator/DEO enter the user information. This includes:

- Membership number
- Photograph
- Roll No.
- Name
- School
- Programme
- Father's name
- Date of birth
- Address
- Telephone
- Email
- Membership date
- Valid up to
- Password

Once the administrator/DEO provides the requested information, the system checks that student membership number is unique. The student is added to the system.

Basic Flow 2: Update a Student

1. The system requests that the administrator/DEO enter the student's membership number.
2. The administrator/DEO enters the student's membership number.

3. The system retrieves and displays the student's information.
4. The administrator/DEO makes the desired changes to the student information. This includes any of the information specified in the **Add a Student** subflow.
5. Once the administrator/DEO updates the necessary information, the system updates the student record with the updated information.

Basic Flow 3: Delete a Student

1. The system requests that the administrator/DEO specify the membership number of the student.
2. The administrator/DEO enters the membership number. The system retrieves and displays the student information.
3. The system prompts the administrator/DEO to confirm the deletion of the student record.
4. The administrator/DEO verifies the deletion.
5. The system deletes the record.

Basic Flow 4: View a Student

1. The system requests that the administrator/DEO specify the membership number.
2. The system retrieves and displays the student information.

Alternative Flows

Alternative Flow 1: Invalid Entry

If in the **Add a Student** or **Update a Student** flow, the actor enters invalid photograph/Roll No./Name/School/Programme/Father's name/Date of birth/Address/Telephone/Email/Membership date/Valid up to/Password or leaves the photograph/Roll No./Name/School/Programme/Father's name/Date of birth/Address/Telephone/Email/Membership date/Valid up to/Password empty, the system displays an appropriate error message. The actor returns to the basic flow and may reenter the invalid entry.

Alternative Flow 2: Student Already Exists

If in the **Add a Student** flow, a student with a specified membership number already exists, the system displays an error message. The administrator returns to the basic flow and may reenter the student information.

Alternative Flow 3: Student Not Found

If in the **Update a Student** or **Delete a Student** or **View a Student** flow, the student information with the specified membership number does not exist, the system displays an error message. The administrator returns to the basic flow and may reenter the membership number.

Alternative Flow 4: Update Cancelled

If in the **Update a Student** flow, the administrator decides not to update the student, the update is cancelled and the **Basic Flow** is restarted at the beginning.

Alternative Flow 5: Delete Cancelled

If in the **Delete a Student** flow, the administrator decides not to delete the student, the delete is cancelled and the **Basic Flow** is restarted at the beginning.

Alternative Flow 6: Deletion Not Allowed If in the **Delete a Student** flow, issue/return/reserve details of the student selected exist, then the system displays an error message. The administrator returns to the basic flow and may reenter the membership number. **Alternative Flow 7: User Exits** This allows the user to exit at any time during the use case. The use case ends.
Special Requirements None
Associated Use Cases Login
B. Validity Checks (i) Only the administrator/DEO will be authorized to access the Student Membership Details module. (ii) Every student will have a unique membership number. (iii) Membership number can only have value from 100 to 5999 digits. (iv) Membership number will not accept alphabets, special characters and blank spaces. (v) Every student will have a unique membership number. (vi) Roll no. cannot be blank. (vii) Length of Roll no. for any user can only be equal to 11 digits. (viii) Roll no. cannot contain alphabets, special characters and blank spaces. (ix) Student name cannot be blank. (x) Length of student name can be of 3 to 50 characters. (xi) Student name will only accept alphabetic characters and blank spaces. (xii) School name cannot be blank. (xiii) Programme name cannot be blank. (xiv) Father's name cannot be blank. (xv) Father's name cannot include special characters and digits, but blank spaces are allowed. (xvi) Father's name can have length 3 to 50 characters. (xvii) Date of birth cannot be blank. (xviii) Address cannot be blank. (xix) Address can have length up to 10 to 200 characters. (xx) Phone cannot be blank. (xxi) Phone cannot include alphabets, special characters and blank spaces. (xxii) Phone can be up to 11 digits. (xxiii) Email cannot be blank. (xxiv) Email can have up to 50 characters. (xxv) Email should contain @ and . characters (xxvi) Email cannot include blank spaces. (xxvii) Password cannot be blank (initially autogenerated of 8 digits).

(xxviii) Password can have length from 4 to 15 characters.
(xxix) Alphabets, digits and hyphen, underscore characters are allowed in password field. However, blank spaces are not allowed.

C. Sequencing Information
None

D. Error Handling/Response to Abnormal Situations
If any of the validations/sequencing flow does not hold true, an appropriate error message will be prompted to the administrator for doing the needful.

3.3 Performance Requirements

Static and dynamic numerical requirements are addressed. Static numerical requirements may include:

(i) The number of terminals to be supported.
(ii) The number of simultaneous users to be supported.
(iii) Amount and type of information to be handled.

Dynamic requirements may include the amount of data to be processed within certain time periods for both normal and peak load conditions, response time for transactions etc. We may specify "75% of transactions shall be processed in less than 2 seconds". Hence, response time for all operations shall be specified. Broadly, it covers those requirements that affect the performance of the software system. For the LMS, we may specify performance requirements as:

3.3 Performance Requirements
(a) Should support at least 8 terminals.
(b) Should support at least 7 users simultaneously.
(c) Should run on 500 MHz, 512 MB RAM machine.
(d) Responses should be within 2 seconds.

3.4 Logical Database Requirements

We may specify logical requirements for any information that is to be placed into a database. We may include:

(i) Frequency of use
(ii) Accessing capabilities
(iii) Data entities and their relationships
(iv) Integrity constraints
(v) Data retention policy

The logical database requirements for the LMS are given in Appendix.

3.5 Design Constraints

If there are any specific design constraints due to some standards, hardware and networking limitations, then those constraints should be specified to help the designers. This may include any deviation from standard (if any) report format, data naming, accounting procedures, audit tracing, etc.

3.6 Software System Attributes

Non-functional requirements such as reliability, availability, security, maintainability, usability, etc. are specified in this subsection. Definitions of some quality attributes are given in Table 3.16.

Table 3.16 Software quality attributes

S. No.	Attribute	Description
1	Reliability	The extent to which a software product performs its intended functions without failure.
2	Correctness	The extent to which the software meets its specifications.
3	Consistency and precision	The extent to which the software is consistent and gives results with precision.
4	Robustness	The extent to which the software tolerates the unexpected problems.
5	Simplicity	The extent to which the software is simple in its operations.
6	Traceability	The extent to which an error is traceable in order to fix it.
7	Usability	The extent of effort required to learn, operate and understand the functions of the software.
8	Accuracy	Meeting specifications with precision.
9	Clarity and accuracy of documentation	The extent to which documents are clearly and accurately written.
10	Conformity of operational environment	The extent to which the software is in conformity of operational environment.
11	Completeness	The extent to which the software has specified functions.
12	Efficiency	The amount of computing resources and code required by the software to perform a function.
13	Testability	The effort required to test the software to ensure that it performs its intended functions.
14	Maintainability	The effort required to locate and fix an error during maintenance phase.
15	Modularity	It is the extent of ease to implement, test, debug and maintain the software.
16	Readability	The extent to which the software is readable in order to understand.
17	Adaptability	The extent to which the software is adaptable to new platforms and technologies.
18	Modifiability	The effort required to modify the software during maintenance phase.
19	Expandability	The extent to which the software is expandable without undesirable side effects.
20	Portability	The effort required to transfer a program from one platform to another platform.

The quality considerations in the LMS are given as follows:

3.6 Software System Attributes

Usability

The application will be user-friendly and easy to operate, and the functions will be easily understandable.

Reliability

The applications will be available to the students throughout the registration period and have a high degree of fault tolerance.

Security

The application will be password protected. Users will have to enter correct login ID and password to access the application.

Maintainability

The application will be designed in a maintainable manner. It will be easy to incorporate new requirements in the individual modules.

Portability

The application will be easily portable on any windows-based system that has SQL Server installed.

IEEE Std 830-1998 has given various ways to organize section 3 and depending on the project, customer's expectations and developers expertise, we may choose one way for the organization of the SRS document.

3.7 Organizing the Specific Requirements

The details are given in the beginning of section 3.

3.8 Additional Comments

We may choose any one organization and the selection is dependent on the developer's expertise, previous experience, customer's expectations, available technologies, etc. Sometimes, more than one organization may also be used for organizing section 3 of the SRS document.

4. Supporting Information

Supporting information makes the SRS easier to use. It may include table of contents, index, appendixes, etc.

3.8 Requirements Change Management

Requirements change management is a new area which has been rapidly adopted by the software industry. As we all know, changes in requirements are inevitable. Many times, we make changes due to factors beyond our control. Every change is difficult and requires a systematic way to handle. Developers keep on getting requests for changes in the requirement before and also after release of the software system. Sometimes, requests for addition of new requirements come, which are equally difficult to handle and incorporate in the present set of requirements. Managing